This shows that irreducible representations of $GL(V)$ which occur in $V^{\otimes n}$ for some n are labeled by Young diagrams with any number of squares but at most $N = \dim V$ rows.

Proposition 5.22.2. *The representation $L_{\lambda+1^N}$ (where $1^N = (1, 1, \ldots, 1) \in \mathbb{Z}^N$) is isomorphic to $L_\lambda \otimes \wedge^N V$.*

Proof. Indeed, $L_\lambda \otimes \wedge^N V \subset V^{\otimes n} \otimes \wedge^N V \subset V^{\otimes n+N}$, and the only component of $V^{\otimes n+N}$ that has the same character as $L_\lambda \otimes \wedge^N V$ is $L_{\lambda+1^N}$. This implies the statement. $\square$

5.23. Algebraic representations of $GL(V)$

Definition 5.23.1. We say that a finite dimensional representation Y of $GL(V)$ is **algebraic** (or **rational**, or **polynomial**) if its matrix elements are polynomial functions of the entries of g, g^{-1}, $g \in GL(V)$ (i.e., belong to $k[g_{ij}][1/\det(g)]$).

Note that subrepresentations, quotients, direct sums, tensor products and duals of algebraic representations are algebraic. For example, $V^{\otimes n}$ and hence all L_λ are algebraic. Also define $L_{\lambda - r\cdot 1^N} := L_\lambda \otimes (\wedge^N V^*)^{\otimes r}$ (this definition makes sense by Proposition 5.22.2). This is also an algebraic representation. Thus we have attached a unique irreducible algebraic representation L_λ of $GL(V) = GL_N$ to any sequence $(\lambda_1, \ldots, \lambda_N)$ of integers (not necessarily positive) such that $\lambda_1 \geq \cdots \geq \lambda_N$. This sequence is called the **highest weight** of L_λ.

Theorem 5.23.2. *(i) Every finite dimensional algebraic representation of $GL(V)$ is completely reducible, and decomposes into summands of the form L_λ (which are pairwise nonisomorphic).*

(ii) (The Peter-Weyl theorem for $GL(V)$) Let R be the algebra of polynomial functions on $GL(V)$. Then as a representation of $GL(V) \times GL(V)$ (with action $(\rho(g,h)\phi)(x) = \phi(g^{-1}xh)$, $g, h, x \in GL(V)$, $\phi \in R$), R decomposes as

$$R = \bigoplus_\lambda L_\lambda^* \otimes L_\lambda,$$

where the summation runs over all λ.